Natural class reasoning in segment deletion rules*

Kyle Gorman[1] and Charles Reiss[2]

[1]CUNY Graduate Center, New York
[2]Concordia University, Montréal

NELS 56, New York University; draft as of November 2025

1 Introduction

Logical Phonology (LP; e.g., Bale et al. 2014, Bale and Reiss 2018, Bale et al. 2020, Reiss 2021, Dabbous et al. 2024, Leduc et al. 2024, Reiss 2024, Gorman and Reiss 2025a,b) is characterized by an insistence on a rigorous definition of the natural classes defining rule targets and environments, and by three novel operators which replace the $\rightarrow$ operator of classical generative phonology. Following these ideas to their logical conclusion has a counterintuitive consequence for the analysis of segment deletion rules, stated below.

(1) Delete The Rich (Reiss 2025): If some—but not all—X's delete in some phonological context Y ___ Z, the X's that delete must be **more richly specified** than those which do not.

After some formal preliminaries (section 2), this prediction is illustrated using deletion processes in Hungarian (section 3) and Turkish (section 4). In Hungarian, there is independent evidence for underlyingly distinct sources of *h*, and these two *h*'s show differential behavior with respect to deletion. In Turkish, positing distinct sources for deleting and non-deleting *k* allows for a unified handling, within the narrow phonology, of apparent exceptions and an atypical form of non-derived environment blocking. Together, these two case studies show that Delete The Rich is not an empirical problem.

2 Formal preliminaries

This section reviews some details of the theory of rules in Logical Phonology. Many other details not strictly relevant to the deletion rules are set aside. Note also that LP itself is a work in progress and the fragment of it presented here is not intended as a complete theory of phonology. Rather, LP is embedded in a network of related assumptions and theories. It is *substance-free*

*We thank Péter Siptár for assistance with the Hungarian data.

in that rejects the notion of markedness in all its senses (e.g., Reiss 2017). Rather than deriving typological tendencies and gaps from markedness, LP takes these patterns to reflect an extra-grammatical *diachronic filter* —as developed in the works of Blevins (2004), Hale (2003, 2007), and Ohala (2003), among others—or from substantive properties of the language acquisition device (e.g., Gorman and Reiss 2025a). It adopts Cognitive Phonetics (e.g., Volenec and Reiss 2018, 2019, 2025) as a theory of the universal phonology-phonetics interface, and thus rejects the hypothesis of language-specific phonetic implementation.

2.1 Features, segments, and natural classes

LP assumes that universal grammar provides a universal innate feature set (e.g., Chomsky and Halle 1965, Reiss and Volenec 2022), and further that these features are binary-valued (i.e., equipollent). A *feature specification* is thus a (possibly empty) set of valued features.

The feature specifications of segments must be *consistent*: if a segment is specified +F, it cannot also be specified −F and vice versa. However, feature specifications do not need to be *complete*: underspecification is permitted. Note that this should not be considered an axiom of LP; rather, completeness is a commonly-adopted axiom which LP happens to lack. LP is thus formally simpler than models which adopt completeness and forbid underspecification.

Segments are consistent feature specifications linked to an X-slot. Some examples are given below; note that the □ symbol in the specification of /I/ has no theoretical status, but is just intended to draw the reader's attention to /I/'s underspecification with respect to High.

(2) Sample segments:

$$/\text{i}/ = \begin{array}{c}\text{X}\\ |\\ \left\{\begin{array}{c}+\textsc{Syllabic}\\ -\textsc{Back}\\ -\textsc{Round}\\ +\textsc{High}\\ -\textsc{Low}\\ +\text{ATR}\end{array}\right\}\end{array} \qquad /\text{e}/ = \begin{array}{c}\text{X}\\ |\\ \left\{\begin{array}{c}+\textsc{Syllabic}\\ -\textsc{Back}\\ -\textsc{Round}\\ -\textsc{High}\\ -\textsc{Low}\\ +\text{ATR}\end{array}\right\}\end{array} \qquad /\text{I}/ = \begin{array}{c}\text{X}\\ |\\ \left\{\begin{array}{c}+\textsc{Syllabic}\\ -\textsc{Back}\\ -\textsc{Round}\\ \square\textsc{High}\\ -\textsc{Low}\\ +\text{ATR}\end{array}\right\}\end{array}$$

Natural classes are also defined by sets of valued features, but following Bale et al. (2020), who formalize a common intuition, they are interpreted as sets of segments. That is, a natural class "matches" (i.e., contains as a member) any segment whose specification is a superset of the natural class's specification. For example, the following shows the intension and extension of the natural class defined by +High.

(3) $\left[+\textsc{High}\right] = \left\{x \mid x \supseteq \left\{+\textsc{High}\right\}\right\} = \{\text{i}, \text{y}, \text{ɨ}, \text{ʉ}, \text{ɯ}, \text{u}, \ldots\}$

For type consistency, curly braces are used to denote segments (and sets in general) and square brackets are used for natural classes. As a notational convenience, a circle, along with a phoneme symbol and square brackets, is also used to abbreviate natural class specifications, as in the example below.

(4) $\left[\text{ⓘ}\right] = \left\{x \mid x \supseteq \{+\textsc{Syllabic}, -\textsc{Back}, -\textsc{Round}, +\textsc{High}, -\textsc{Low}, +\text{ATR}\}\right\} = \{\text{i}\}$

LP assumes that targets and triggers (i.e., contexts or environments) of individual rules are intensionally defined by natural classes.[1] This does not entail that all phonological processes, informally speaking, have an extension which is itself a natural class. For example, a rule might extensionally target a set of segments that is smaller than its intensionally defined natural class because it is bled by an earlier rule removing potential targets; see Reiss and Volenec 2022 (29f.) for a concrete example involving English regular plurals. This does not exhaust the possible divergences between the intension and extension, as seen below.

Of course, some natural classes, like (4), have a single-phoneme extension, and can thus serve as targets and triggers under this assumption. However, one consequence of this definition of natural classes—noted by Lees (1961:12–14) and Lightner (1971:236), among others—is that one cannot specify a natural class that contains underspecified segment, say /I/, but does not also contain all segments whose specification is a proper superset of that segment, say /i, e/. This is exemplified and schematized below.

(5) $\left[\text{Ⓘ}\right] = \left\{x \mid x \supseteq \{+\textsc{Syllabic}, -\textsc{Back}, -\textsc{Round}, -\textsc{Low}, +\text{ATR}\}\right\} = \{\text{i}, \text{e}, \text{I}\}$

(6) Specificity: Let f, g be segments (feature specifications) such that $f \subset g$. Then there is no natural class containing f but excluding g.

As will be seen, Delete The Rich is a direct consequence of Specificity.

2.2 Rules

LP replaces the traditional $\rightarrow$ operator with three operators defined below. The *subtraction* operator $\setminus$ removes valued features from a segment, and thus corresponds to set difference.

(7) Subtraction: $A \setminus B = \{c\text{F} \mid c\text{F} \in A \wedge c\text{F} \notin B\}$

The second, the *unification* operator $\sqcup$, adapted from syntactic theory (e.g., Shieber 1986), adds a valued feature to a segment, but has no effect if the target is already specified for that feature.

(8) Unification: $A \sqcup B = A \cup \{c\text{F} \mid c\text{F} \in B \wedge -c\text{F} \notin A\}$

In other words, $A \sqcup B$ contains all features of A as well as any features of B which do not conflict with those in A. Thus unification only adds valued features when the target (the first argument) is underspecified with respect to the features in the change, as illustrated below. Unification is thus a solution to Specificity, i.e., the problem of targeting underspecified segments, and corresponds to so-called *feature-filling* (or *structure-building*) processes. A toy example is given below.

[1] In contrast, some theorists assume phonological processes merely tend to be defined by natural classes; for example, Hayes (2009:43) writes that "in most instances, the segments that undergo a rule or appear in the environment of a rule form a natural class"; and Rasin et al. (2021) develop a model in which rule target and triggers may also be defined by sets of segments not necessarily forming a natural class.

(9) Yield of unification:

a. “Feature addition”: $\left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\end{matrix}\right\} \sqcup \left\{-\textsc{High}\right\} \rightsquigarrow \left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\\-\textsc{High}\end{matrix}\right\}$

b. “Vacuous application”: $\left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\\-\textsc{High}\end{matrix}\right\} \sqcup \left\{-\textsc{High}\right\} \rightsquigarrow \left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\\-\textsc{High}\end{matrix}\right\}$

c. “Unification failure”: $\left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\\\underline{+\textsc{High}}\end{matrix}\right\} \sqcup \left\{\underline{-\textsc{High}}\right\} \rightsquigarrow \left\{\begin{matrix}-\textsc{Back}\\-\textsc{Round}\\\underline{+\textsc{High}}\end{matrix}\right\}$

These two operators are used to define rules where the target is a natural class, the change is a feature specification, and the rule’s context for application are optionally specified using natural class-defined triggers to the left and/or right.[2] This gives rise to the following rule schemata.

(10) Subtraction schema: $[\ldots] \setminus \{\ldots\} \;/\; [\ldots] \_\_ [\ldots]$

(11) Unification schema: $[\ldots] \sqcup \{\ldots\} \;/\; [\ldots] \_\_ [\ldots]$

Following prior suggestions (e.g., Poser 1982, Mester and Itô 1989, Inkelas and Cho 1993, Siptár and Törkenczy 2000), so-called *feature-changing* (or *structure-changing*) processes, such as final devoicing, are thus derived from a subtraction rule followed by a unification rule.[3]

For example, consider final devoicing in Russian, a feature-changing process.

(12) Russian final devoicing:

	nom. sg.	gen. sg.	
a.	t͡svʲet	t͡svʲeta	‘color’
b.	prut	pruda	‘pond’

In LP, final devoicing derives from a subtraction rule removing the Voice specification of final obstruents, followed by a unification rule adding −Voice. The unification can be written more broadly because feature addition will only obtain for segments targeted by the subtraction rule.

(13) $[-\textsc{Sonorant}] \setminus \{+\textsc{Voice}\} \;/\; \_\_ \%$

(14) $[\;] \sqcup \{-\textsc{Voice}\}$

The ordered application of these two rules is illustrated using a segment mapping diagram (SMD; see Bale and Reiss 2018:§19), with /D/ symbolizing an alveolar stop not specified for Voice.

[2]For ease of exposition we naïvely assume that targets and triggers are always local to each other; see Dabbous et al. 2024 for a detailed theory of locality and long-distance effects in LP. Furthermore, in lieu of a complete formalization of suprasegmental conditioning of rules, we refer to word boundaries and syllable edges as needed.

[3]This does not exhaust the options available in LP. For example, Gorman and Reiss (2025b:§5) decompose a metaphony process in Cervara into a sequence consisting of one unification, one subtraction, and three unifications.

(15) Russian final devoicing SMD:

$$\begin{array}{rccc}
\text{UR:} & \text{t} & & \text{d} \\
 & \downarrow & \swarrow & \downarrow \\
\text{Rule (13):} & \text{t} & \text{D} & \text{d} \\
 & \downarrow & \swarrow & \downarrow \\
\text{Rule (14):} & \text{t} & & \text{d} \\
 & \downarrow & & \downarrow \\
\text{SR:} & \text{t} & & \text{d}
\end{array}$$

LP's ability to generate feature-changing processes via interacting rules means that it does away with the feature-filling/feature-changing rule diacritics used in much prior work (e.g., Kiparsky 1985, Inkelas and Cho 1993, Buckley 1994, Inkelas 1995).

The third operator in LP is the *segment* operator $\mapsto$, which is combined with a null symbol ϵ to implement insertion and deletion of entire segments.[4] Schemas for insertion and deletion rules are given below.

(16) Insertion schema:

$$\epsilon \;\mapsto\; \begin{array}{c}\mathrm{X}\\ | \\ \{\ldots\}\end{array} \;/\; [\ldots] \;\_\_\; [\ldots]$$

(17) Deletion schema:

$$\begin{array}{c}\mathrm{X}\\ | \\ [\ldots]\end{array} \;\mapsto\; \epsilon \;/\; [\ldots] \;\_\_\; [\ldots]$$

In insertion rules, the X is linked to a feature bundle and a specific segment is inserted. In deletion rules, however, the X is linked to a natural class and any segment matching the natural class is deleted. Because of this, targeting in deletion rules conforms Specificity. In unification rules, this Specificity effect is concealed by vacuous application and unification failure, but there is no such escape hatch for deletion rules.[5] That is, a unification rule of the form $[\text{I}] \sqcup \{+\textsc{High}\}$ has an observable effect only on /I/, whereas a deletion rule targeting $[\text{I}]$ targets /i, e, I/ both intensionally and extensionally, just as predicted by Delete The Rich.

Delete The Rich follows directly from the LP formalism, but runs contrary to a common functionalist intuition that less richly specified segments are "weaker" and thus are preferential targets for deletion (e.g., van Oostendorp 2003, Silverman 2011; see Reiss 2025 for discussion). In LP, a substance-free theory, however, this intuition is an example of *substance abuse* (in the sense of Hale and Reiss 2008:§8), and more generally an example of "the mistake of attributing to a mental representation the properties of what it represents" (Pylyshyn 2003:8)

[4]Note that LP does not model segment deletion via subtraction. Subtraction removes features, not segments, and segmental underspecification is representationally distinct from segmental absence.

[5]An anonymous reviewer asks whether one could decompose segment deletion into subtraction followed by deletion of the X. However, Delete The Rich entails the second rule—deleting only those X's linked to a null feature bundle—is unstatable in LP.

2.3 Epistemic boundedness

We now use these tools to reanalyze cases of segment deletion with putative exceptions under the hypothesis that segments which behave differently in the same context are underlyingly distinct. Nearly all phonologists work under the–usually implicit—premise that it is preferable to provide an empirically adequate analysis within the "narrow" phonology rules rather than using various forms of morpheme-specific exceptionality, cophonologies, morphophonological rules conditioned on particular lists of morphemes, morphosyntactic rules conditioned on morphosyntactic category or morphemic structure, or suppletion of stems or affixes. This premise is central to the theory of phonology and its interfaces, and it has long been recognized that what we are calling narrow phonology is necessary to account for the characteristic *systematicity* and *productivity* (in the sense of Kenstowicz and Kisseberth 1979:§2) of many sound patterns. Despite this, there is little in the way of mechanistic explanation for how the preference for narrow phonology is expressed during language acquisition. As an answer to this question we hypothesize that children are *epistemically bounded* (in the sense of Fodor 1980:33f.) to pursue narrow phonological analyses, resorting to these other mechanisms only when the (narrowly-defined) phonological module fails to provide an empirically-adequate solution (see also Gorman and Reiss 2025b:§2.4).

LP is a theory of the narrow phonological module, and it thus follows that the child so endowed with such a module would make use of its features in analyzing putative exceptions to deletion. LP permits featural underspecification; Delete The Rich provides a mechanism by which segments can be exempted from deletion rules via underspecification; and the feature-filling effect of unification provides a mechanism by which these underspecified segments can be made surface-identical to deleting segments when deletion does not obtain. We thus predict the child will make assiduous use of underspecification as proposed in the following analyses.

3 Hungarian *h*-deletion

Some Hungarian *h*'s delete in the coda, but other *h*'s are realized as coda [x].

(18) Hungarian *h*'s (Siptár and Törkenczy 2000:§8.2.2):

	nom. sg.	abl. sg.	nom. pl.	
a.	t͡ʃɛ	t͡ʃɛ-tøːl	t͡ʃɛɦ̱-ɛk	'Czech'
	ju	ju-toːl	juɦ̱-ok	'sheep'
	meː	meː-tøːl	meːɦ̱-ɛk	'bee'
b.	dox̱	dox̱-toːl	doɦ̱-ok	'musty smell'
	ɛunux̱	ɛunux̱-toːl	ɛunuɦ̱-ok	'eunuch'
	meːx̱	meːx̱-tøːl	meːɦ̱-ɛk	'womb'

Siptár and Törkenczy, henceforth S&T, take (18a) to involve exceptional deletion, and derive onset [ɦ] in (18b), the "systematic pattern", from underlying /x/.

> We claim that in present-day ECH [Educated Colloquial Hungarian] it is the *doh* [(18b)] type that is the systematic pattern…rather than the *cseh* [(18a)] type, which we suggest is not phonological (anymore) and is best considered as suppletive allomorphy. (S&T:275)

In contrast, we hypothesize that roots showing the [ɦ]–zero alternation end in /h/, and those that show the [ɦ]–[x] alternation end in /H/, a segment underspecified for Consonantal. Partial feature specification for /h, H/ and their surface correspondents [ɦ, x] are given below.

(19) Partial feature specifications:

	/h/	[ɦ]	/H/	[x]
Consonantal	−	−		+
Dorsal				+
Voice	−	+	−	−

The following deletion rule targets coda /h/.

(20) *h*-deletion:

$$\begin{array}{c} \mathrm{X} \\ | \\ \left[\text{h}\right] \end{array} \mapsto \epsilon \;/\; \_\_\,]_\sigma$$

/H/ is unaffected by this rule. Indeed, Delete The Rich entails that no deletion rule could derive the [h]–zero alternation from underlying /H/ without also deleting coda /h/.

The following rules derive the surface forms of /H/. We assume that /H/ is the only segment not specified for the relevant features at this point of the derivation, so these rules can be stated broadly. The first rule maps /H/ ⇝ [x] in coda position and the second maps it to [h] elsewhere.

(21) $[\ ] \sqcup \left\{\begin{matrix}+\textsc{Consonantal} \\ +\textsc{Dorsal}\end{matrix}\right\} \;/\; \_\_\,]_\sigma$

(22) $[\ ] \sqcup \{-\textsc{Consonantal}\}$

Finally, [h]—whether present underlyingly or derived from /H/ by (22)—is voiced between two vowels or between a sonorant consonant and a vowel. A few examples are given below.

(23)	a.	/t͡ʃɛh-Ek/	⇝	[t͡ʃɛɦ-ɛk]	‘Czechs’
	b.	/doH-Ek/	⇝	[doɦ-ok]	‘musty smells’
	c.	/koɲhɔ/	⇝	[koɲɦɔ]	‘kitchen’

This feature-changing process is implemented by subtraction followed by unification.

(24) $\left[\text{h}\right] \setminus \{-\textsc{Voice}\} \;/\; [+\textsc{Sonorant}] \;\_\_\; [+\textsc{Sonorant}]$

(25) $[\ ] \sqcup \{+\textsc{Voice}\}$

As stated, the above rules instantiate three critical orderings: (20) must apply before (21), which would counterfactually bleed it; (21) must apply before (22) because the latter is stated more broadly, and (24) must apply before (25) to derive the observed feature-changing effect. Sample derivations are given below.

(26)

	‘Czech’	(abl. sg.)	(nom. pl.)	‘musty smell’	(abl. sg.)	(nom. pl.)
UR:	t͡ʃɛh	t͡ʃɛh-tO:l	t͡ʃɛh-Ek	doH	doH-tO:l	doH-Ek
Harmony:		t͡ʃɛhtø:l	t͡ʃɛhɛk		doHto:l	doHok
Rule (20):	t͡ʃɛ	t͡ʃɛtø:l				
Rule (21):				dox	doxto:l	
Rule (22):						dohok
Rules (24–25):			t͡ʃɛɦɛk			doɦok
SR:	t͡ʃɛ	t͡ʃɛtø:l	t͡ʃɛɦɛk	dox	doxto:l	doɦok

The analysis sketched thus far locates the distinct behavior of the two *h*’s fully within the narrow phonology, without recourse to the exceptional suppletion proposed by S&T.

Some confirmatory evidence for the proposed specification of /H/ comes from its behavior with respect to voice assimilation. Adjacent consonants undergo regressive voice assimilation.[6] However, /H/ is not a target for voice assimilation, as can be seen from (27c); e.g., *[me:ɣ̲-bɛn].

(27) Regressive voice assimilation (S&T:§4.1.1; Péter Siptár, p.c.):

	nom. sg.	abl. sg.	iness. sg.	all. sg.	
a.	kɔlɔp	kɔlɔp-to:l	kɔlɔb̲-bɔn	kɔlɔp-hoz	‘hat’
	se:f	se:f-tø:l	se:v̲-bɛn	se:f-hɛz	‘safe’
	re:s	re:s-tø:l	re:z̲-bɛn	re:s-hɛz	‘part’
	ʒa:k	ʒa:k-to:l	ʒa:g̲-bɔn	ʒa:k-hoz	‘sack’
b.	rɔb	rɔp̲-to:l	rɔb-bɔn	rɔp̲-hoz	‘captive’
	ka:d	ka:t̲-to:l	ka:d-bɔn	ka:t̲-hoz	‘bathtub’
	mɛlɛg	mɛlɛk̲-tø:l	mɛlɛg-bɛn	mɛlɛk̲-hɛz	‘warmth’
c.	dox	dox-to:l	dox-bɔn	dox-xoz	‘musty smell’
	me:x	me:x-tø:l	me:x-bɛn	me:x-xez	‘womb’

However, /H/ is a trigger of voice assimilation, devoicing preceding obstruents; e.g., /ɔd-Hɔt/ ⇝ [ɔt̲-hɔt] ‘he may give’. This behavior is predicted by the feature specification given above: /H/ is not a target for voice assimilation because it is not +CONSONANTAL, but it is a trigger—i.e., it is *catalytic* in the sense of Gorman and Reiss 2025b—of voice assimilation because it is specified for VOICE. Voice assimilation can thus be stated by the following rules (see Reiss 2021:§6.2).

(28) $$\begin{bmatrix} -\textsc{Sonorant} \\ +\textsc{Consonantal} \end{bmatrix} \setminus \{\alpha\textsc{Voice}\} \; / \; \_\_ \begin{bmatrix} -\textsc{Sonorant} \\ -\alpha\textsc{Voice} \end{bmatrix}$$

(29) $$\begin{bmatrix} -\textsc{Sonorant} \end{bmatrix} \sqcup \{\alpha\textsc{Voice}\} \; / \; \_\_ \begin{bmatrix} -\textsc{Sonorant} \\ \alpha\textsc{Voice} \end{bmatrix}$$

While this data seems to confirm the specification of /H/ proposed above, a few questions remain; for example, what accounts for geminate in [dox-xoz, me:x-xez]?

[6] The behavior of *v* poses a related set of difficulties for the analysis of voice assimilation. Using similar arguments to the ones given above for *h*, Reiss (2021, 2024) argues there are also two underlyingly distinct sources for *v*.

4 Turkish *k*-deletion

Some Turkish *k*’s delete intervocalically (all data from Inkelas et al. 2000 unless otherwise noted).

		nom. sg.	def. acc. sg.	1sg. poss.	
(30)	a.	konuk̲	konu-u	konu-u̲m	‘guest’
		tyfek̲	tyfe-i	tyfe-i̲m	‘rifle’
	b.	sokak̲	soka-ɯ	soka-ɯ̲m	‘street’
		tykyɾyk̲	tykyɾy-y	tykyɾy-y̲m	‘spit’
	c.	fiki̲ɾ	fikɾ-i	fikɾ-i̲m	‘idea’
		hyky̲m	hykm-y	hykm-y̲m	‘judgment’
	d.	hukuk	hukuk-u	hukuk-u̲m	‘law’
		meɾak	meɾak-ɯ	meɾak-ɯ̲m	‘curiosity’
	e.	kedi	kedi-j̲i	kedi-m	‘cat’
		kod͡ʒa	kod͡ʒa-j̲ɯ	kod͡ʒa-m	‘husband’
	f.	joɾum	joɾum-u	joɾum-u̲m	‘comment’
		sanat	sanat-ɯ	sanat-ɯ̲m	‘art’

The most common pattern, affecting hundreds of nouns, is shown in (30a): a root-final *k* preceded by a root vowel deletes after vowel-initial suffixes. Two details should be briefly mentioned. First, some root-final *g*’s also delete in this context (e.g., [katalog]–[katalo-a] ‘catalog’), but this only affects a few loanwords, and according to Clements and Sezer (1982:253), speakers tend to pronounce these orthographic root-final *g*’s as [k]. Thus for ease of exposition, the analysis below focuses solely on the behavior of *k*. Secondly, there are a few vowel-initial suffixes which never trigger *k*-deletion; Inkelas and Orgun (1995:768) claims these suffixes all belong to “Level 2” morphological stratum in which deletion of *k* is expected to be inactive for independent reasons.

In (30b) root-final intervocalic *k* undergoes deletion but root-internal intervocalic *k* is preserved throughout the paradigm. At first glance, this pattern suggests deletion is subject to *non-derived environment blocking* (NDEB). The notion of *derived environment* is usually defined disjunctively: an environment is said to be derived if it is the result either of morpheme concatenation or other phonological processes. (30c) however shows that *k* does not delete in the latter type of derived environment. These roots underlyingly end in two consonants and undergo epenthesis of a harmonic +High vowel in the nom. sg., when this consonant cluster is word-final (see Inkelas 2009:395f. for details). When the first of the two root-final consonants is *k*, epenthesis renders it intervocalic in the nom. sg., yet the *k* does not delete. Thus deletion of *k* does not conform fully to NDEB as traditionally understood (cf. Inkelas 2000, 2009, 2014:§8.8).

(30e) shows the behavior of vowel-final roots, suggesting that the harmonic +High vowel in the 1sg. poss. suffix is epenthetic but the featurally-identical vowel in the def. acc. sg. is part of the suffix. Finally (30f) shows that non-dorsal consonants do not delete intervocalically.

Although Inkelas is skeptical that *k*-deletion is an instance of NDEB, she also doubts whether the differential behavior of various velars can be handled by the underspecification logic she and colleagues propose for feature-filling processes (e.g., Inkelas and Cho 1993, Kiparsky 1993, Buckley 1994, Inkelas 1995, Inkelas et al. 1997) under the rubric of *inalterability as prespecification*.

> Velar Deletion deletes whole segments. One might attempt various strategies for featural prespecification of nonalternating /k, g/ and underspecification of alternating /k,

g/, but since the alternation itself is not featural, these attempts would be *far-fetched and uninsightful*. (Inkelas 2000:25, emphasis ours)

Yet it is trivial to extend Delete The Rich reasoning applied to Hungarian *h*'s to Turkish *k*'s: *k*'s that delete must be more richly specified than *k*'s that do not. For concreteness, we hypothesize that non-deleting [k] is /K/, a segment underspecified for Dorsal, and deleting [k] is the fully specified /k/. The following rules are then responsible for the observed patterns. The 1sg. poss. suffix /-m/ triggers epenthesis of a harmonizing +High vowel after a word-final consonant.[7]

(31) Epenthesis:

$$\epsilon \mapsto \begin{Bmatrix} \overset{\displaystyle\text{X}}{\big|} \\ -\text{Syllabic} \\ +\text{High} \end{Bmatrix} \;/\; \big[-\text{Syllabic}\big] \;\_\_\; \big[-\text{Syllabic}\big] \;\%$$

In (30ab), the epenthetic vowel also appears after a vowel that becomes root-final due to velar deletion, indicating that epenthesis precedes—and is counterbled by—*k*-deletion. This rule, stated below, does not affect underspecified /K/.

(32) *k*-deletion:

$$\overset{\displaystyle\text{X}}{\underset{\displaystyle\big[\,\text{k}\,\big]}{\big|}} \mapsto \epsilon \;/\; \big[+\text{Syllabic}\big] \;\_\_\; \big[+\text{Syllabic}\big]$$

A final unification rule specifies the remaining /K/s as +Dorsal; other segments are unaffected, under they assumption that they are already specified as either +Dorsal or −Dorsal; alternatively, we can narrow the target class by specifying it as [K].

(33) $[\;] \sqcup \{+\text{Dorsal}\}$

Sample derivations are provided below.

(34)

	'street'	'my street'	'curiosity'	'my curiosity'
UR:	soKak	soKak-m	meraK	meraK-m
Rule (31) + Harmony:		soKakɯm		meraKɯm
Rule (32):		soKaɯm		
Rule (33):	sokak	sokaɯm	merak	merakɯm
SR:	sokak	sokaɯm	merak	merakɯm

[7] This rule is likely stated too broadly, because Turkish permits certain word-final consonant clusters (see Clements and Sezer 1982:§3.5), but the exact generalizations are not settled.

Inkelas and Orgun (1995), among others, claim that monosyllables do not undergo *k*-deletion; e.g., [kœk]–[kœkym] '(my) root'. They appeal to *ad hoc* notions—Bimoraic Minimality and Final Consonant Invisibility—to exempt velars in monosyllables from *k*-deletion. However, there are still some monosyllables which do undergo *k*-deletion.

		nom. sg.	1sg. poss.		
(35)	a.	ʧok̲	ʧo-um	'a lot'	
		gœk̲	gœ-ym	'sky'	
	b.	brik̲	bri-im	'brik (Tunisian pastry)'	(UniMorph; Kirov et al. 2018)
		fɾak̲	fɾa-ɯm	'tailcoat'	
		plak̲	pla-ɯm	'vinyl record'	

While these are analytically-challenging exceptions for Inkelas and Orgun, they pose no issues for the analysis given here, as shown in the sample derivations below.

		'my root'	'my sky'
(36)	UR:	kœK-m	gœk-m
	Rule (31) + Harmony:	kœKym	gœkym
	Rule (32):		gœym
	Rule (33):	kœkym	
	SR:	kœkym	gœym

Under this analysis, /k/ never occurs intervocalically within roots, but this gap is of no great interest, since in the grammar of Turkish we propose a root-internal intervocalic /k/ could never appear on the surface in the first place (cf. Prince and Smolensky 1993:209).

It is thus possible to salvage the underspecification analysis of *k*-deletion for which Inkelas saw little promise. Indeed, the analysis given here demands no new ontological commitments from LP, expressing all the generalizations within the narrow phonology. It is fully modular in the sense that it makes no reference to morphemic structure, either as a locus for exceptionality or as a condition for applications in derived environments.

It was assumed above that non-deleting *k*'s are underspecified for Dorsal, but alternatively, consider the possibility that non-deleting velars are just prosodically underspecified, e.g., in the sense that they lack an underlying X-slot, but are fully specified, just like velars that delete.

(37) Schematic URs for 'street' and 'law':

```
a. X  X     X  X     b. X  X     X
   |  |     |  |        |  |     |
   s  o  k  a  k        h  u  k  u  k
```

The following schematic derivations then show the operation of Epenthesis, Harmony, *k*-deletion, and a rule or rules associating unassociated segments with X's in lieu of (33).

(38) Epenthesis + Harmony:

```
X X   X X X       X X   X X X X
| |   | | |   ⇝   | |   | | | |
s o k a k m       s o k a k ɯ m
```

(39) *k*-deletion + default association:

```
X X   X X X X       X X X X X X
| |   | | | |   ⇝   | | | | | |
s o k a k ɯ m       s o k a ɯ m
```

This approach is consistent with the spirit of Delete The Rich, because *k*-deletion cannot target a segment not linked to an X-slot, but only up to a point: default association needs to be able to target unassociated segments. This apparent violation of Delete The Rich might be taken to reflect default association's status as an "automatic" convention, applying late in the derivation independent of the featural content, but further theorizing is necessary. In other words, it remains to be seen whether there is any reason to favor prosodic versus featural underspecification for the analysis of Turkish.

5 Conclusions

Under the LP conception of segment deletion rules, the deletion of Hungarian *h*'s and Turkish *k*'s are located entirely within the narrow phonology, without reference to morphemic structure. The resulting phonological rules are exceptionless. LP's admittance of underspecification and Delete The Rich follow from LP's minimal ontological commitments—specifically, traditional natural class reasoning made explicit—not stipulations of the theory. This ontological minimality is consonant with the pursuit of minimalist explanatory models in other sciences:

> The simpler the assumptions, the deeper the explanation. (N. Chomsky, "Fundamental Issues in Linguistics", 4/10/2019)[8]

Kiparsky (1993) recognizes that NDEB is a problematic notion and suggests that it would be desirable to reanalyze all putative NDEB effects using underspecification. Delete The Rich suggests a means to do so for NDEB effects involving segment deletion, and the LP framework is sufficiently flexible to handle the atypical, "exceptional" deletion pattern in Turkish. Of course, it remains to be seen whether all putative NDEB effects can be reanalyzed in a similar fashion.

References

Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: A formal analysis of feature-changing and feature-filling rules. *Lingua* 148:240–253.

[8] `https://youtu.be/r514RhgISv0?si=fbJ56Vo-R5oAHaRn&t=2940`

Bale, Alan, and Charles Reiss. 2018. *Phonology: A Formal Introduction*. MIT Press.

Bale, Alan, Charles Reiss, and David Ta-Chun Shen. 2020. Sets, rules and natural classes: {} vs. [ ]. *Loquens* 6:e06.

Blevins, Juliette. 2004. *Evolutionary Phonology*. Cambridge University Press.

Buckley, Eugene. 1994. Prespecification of default features: The two /i/'s of Kashaya. In *NELS 24: Proceedings of Twenty-Fourth Annual Meeting of the North East Linguistic Society*, 17–30.

Chomsky, Noam, and Morris Halle. 1965. Some controversial questions in phonological theory. *Journal of Linguistics* 1:97–138.

Clements, G. N., and Engin Sezer. 1982. Vowel and consonant disharmony in Turkish. In *The Structure of Phonological Representations*, ed. Harry van der Hulst and Norval Smith, 213–254. Foris.

Dabbous, Rim, Marjorie Leduc, Charles Reiss, and David Ta-Chun Shen. 2024. Locality is epiphenomenal: Adjacency is opaqueness. In *Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.

Fodor, Jerry. 1980. Reply to Putnam. In *Language and Learning: The Debate between Jean Piaget and Noam Chomsky*, ed. Massimo Piattelli-Palmarini, 325–334. Harvard University Press.

Gorman, Kyle, and Charles Reiss. 2025a. How not to acquire exchange processes in Logical Phonology. In *Proceedings of the 2025 annual conference of the Canadian Linguistic Association*, in press.

Gorman, Kyle, and Charles Reiss. 2025b. Metaphony in Substance Free Logical Phonology. *Phonology* in press.

Hale, Mark. 2003. Neogrammarian sound change. In *The Handbook of Historical Linguistics*, ed. Brian D. Joseph and Richard D. Janda, 343–368. Blackwell.

Hale, Mark. 2007. *Historical Linguistics: Theory and Method*. Blackwell.

Hale, Mark, and Charles Reiss. 2008. *The Phonological Enterprise*. Oxford University Press.

Hayes, Bruce. 2009. *Introductory Phonology*. Blackwell.

Inkelas, Sharon. 1995. The consequences of optimization for underspecification. In *Proceedings of the North East Linguistic Society 25*, 287–302.

Inkelas, Sharon. 2000. Phonotactic blocking through structural immunity. In *Lexicon in Focus*, ed. Barbara Stiebels and Dieter Wunderlich, 7–40. De Gruyter.

Inkelas, Sharon. 2009. Another look at velar deletion in Turkish, with special attention to the derived environment condition. *UC Berkeley Phonology Lab Annual Report* 5:387–403.

Inkelas, Sharon. 2014. *The Interplay of Morphology and Phonology*. Oxford University Press.

Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69:529–574.

Inkelas, Sharon, Aylin Küntay, C. Orhan Orgun, and Ronald Sprouse. 2000. Turkish Electronic Living Lexicon (TELL). *Turkic Languages* 4:253–275.

Inkelas, Sharon, and Cemil Orhan Orgun. 1995. Level ordering and economy in the lexical phonology of Turkish. *Language* 71:763–793.

Inkelas, Sharon, Orhan Orgun, and Cheryl Zoll. 1997. The implications of lexical exceptions for the nature of grammar. In *Derivations and Constraints in Phonology*, ed. Iggy Roca, 393–418. Oxford University Press.

Kenstowicz, Michael, and Charles Kisseberth. 1979. *Generative Phonology: Description and Theory*. Academic Press.

Kiparsky, Paul. 1985. Some consequences of Lexical Phonology. *Phonology Yearbook* 2:85–138.

Kiparsky, Paul. 1993. Blocking in non-derived enviroments. In *Studies in Lexical Phonology*, ed. Sharon Hargus and Ellen Kaisse, 277–313. Academic Press.

Kirov, Christo, Ryan Cotterell, John Sylak-Glassman, Géraldine Walther, Ekaterina Vylomova, Patrick Xia, ..., and Mans Hulden. 2018. UniMorph 2.0: Universal morphology. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, 1868–1873.

Leduc, Marjorie, Charles Reiss, and Veno Volenec. 2024. Votic vowel harmony in Substance Free Logical Phonology. In *Oxford Handbook of Vowel Harmony*, ed. Harry van der Hulst and Nancy A. Ritter, 425–436. Oxford University Press.

Lees, Robert B. 1961. *The Phonology of Modern Standard Turkish*. Indiana University Publications.

Lightner, Theodore M. 1971. On Swadesh and Voegelin's 'A Problem in Phonological Alternation'. *International Journal of American Linguistics* 37:227–237.

Mester, Armin, and Junko Itô. 1989. Feature predictability and underspecification: Palatal prosody in Japanese mimetics. *Language* 65:258–293.

Ohala, John J. 2003. Phonetics and historical phonology. In *The Handbook of Historical Linguistics*, ed. Brian D. Joseph and Richard D. Janda, 669–686. Blackwell.

van Oostendorp, Marc. 2003. Schwa in phonological theory. In *The Second Glot International State-of-the-Article Book: The Latest in Linguistics*, ed. Lisa Cheng and Rint Sybesma, 431–461. De Gruyter Mouton.

Poser, William. 1982. Phonological representation and action at-a-distance. In *The Structure of Phonological Representations*, ed. Harry van der Hulst and Norval Smith, 121–58. Foris.

Prince, Alan, and Paul Smolensky. 1993. Optimality Theory: Constraint interaction in generative grammar. Technical report TR-2 and CU-CS-696-93. Published by MIT Press, 2004.

Pylyshyn, Zenon W. 2003. *Seeing and Visualizing: It's Not What You Think*. MIT Press.

Rasin, Ezer, Iddo Berger, Nur Lan, Shefi, and Roni Katzir. 2021. Approaching explanatory adequacy in phonology using Minimum Description Length. *Journal of Language Modelling* 9:17–66.

Reiss, Charles. 2017. Substance free phonology. In *The Routledge Handbook of Phonological Theory*, ed. S. J. Hannahs and Anna R.K. Bosch, 425–452. Routledge.

Reiss, Charles. 2021. Towards a complete Logical Phonology model of intrasegmental changes. *Glossa* 6:107.

Reiss, Charles. 2024. Specificity in rule targets and triggers: Two v's in Hungarian. In *Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.

Reiss, Charles. 2025. Delete the rich: On the non-existence of 'weak' schwa deletion. Ms. LOA-006. URL: `https://lingbuzz.net/lingbuzz/008747`.

Reiss, Charles, and Veno Volenec. 2022. Conquer primal fear: Phonological features are innate and substance free. *Canadian Journal of Linguistics* 67:581–610.

Shieber, Stuart. 1986. *An Introduction to Unification-Based Approaches to Grammar*. CSLI Publications.

Silverman, Daniel. 2011. Schwa. In *Blackwell Companion to Phonology*, ed. Marc van Oostendorp, Colin J. Ewen, Beth Hume, and Keren Rice, chapter 26. Wiley Online Library.

Siptár, Péter, and Miklós Törkenczy. 2000. *The Phonology of Hungarian*. Oxford University Press.

Volenec, Veno, and Charles Reiss. 2018. Cognitive phonetics: The transduction of distinctive features at the phonology-phonetics interface. *Biolinguistics* 11:251–294.

Volenec, Veno, and Charles Reiss. 2019. The intervocalic palatal glide in cognitive phonetics. In *Proceedings of the Forty-Ninth Meeting of the North East Linguistics Society*, 255–264.

Volenec, Veno, and Charles Reiss. 2025. Cognitive phonetics: The universal phonology-phonetics interface. In *Cambridge Handbook of Linguistic Interfaces*, ed. Antonio Fábregas, Laia Mayol, and Yanina Prystauka, to appear. Cambridge University Press.